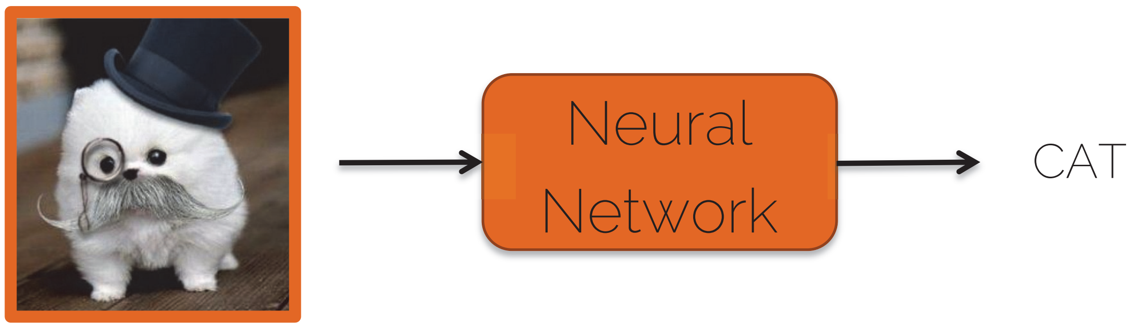
Siamese Neural Networks and Similarity Learning

What can ML do for us?

- Classification problem

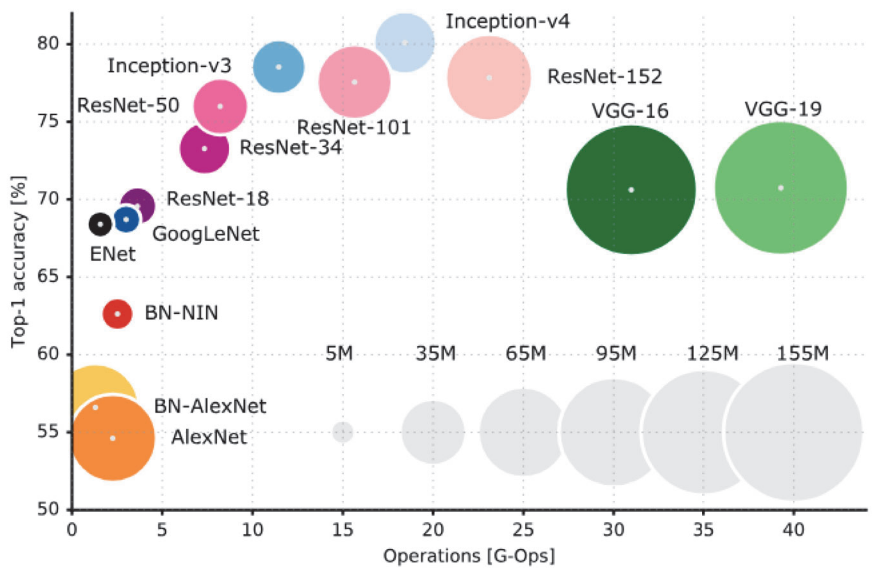


What can ML do for us?

- Classification problem on ImageNet with thousands of categories

What can ML do for us?

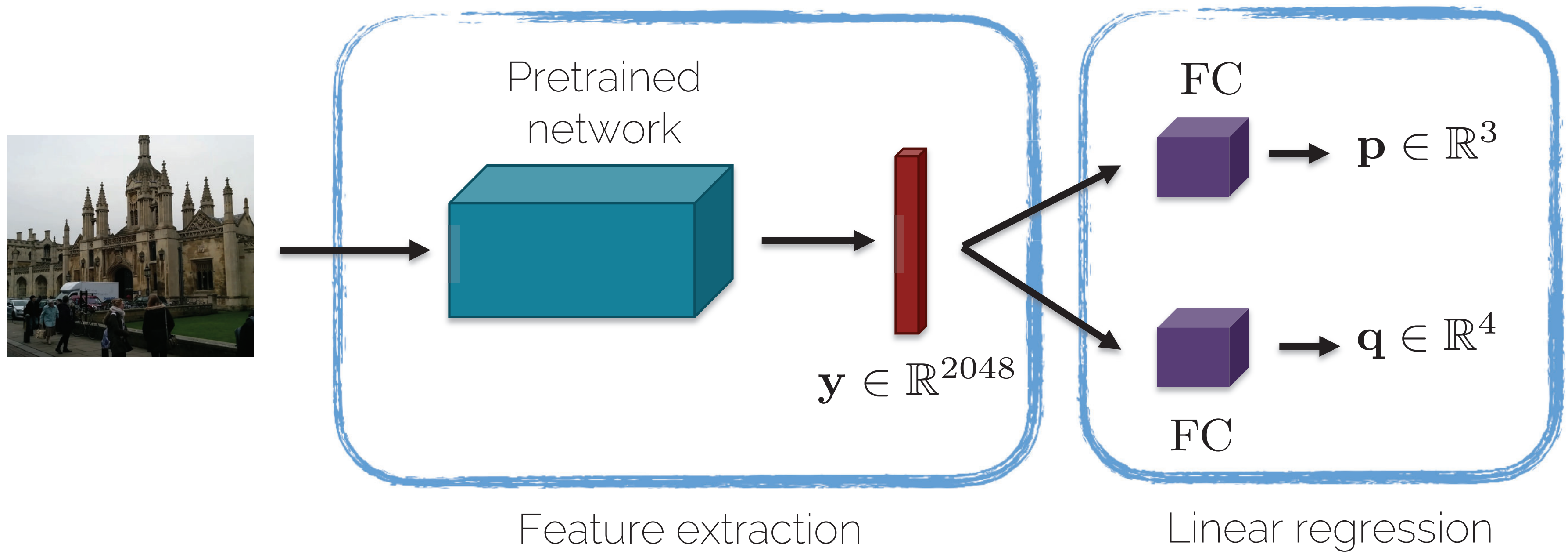
- Performance on ImageNet
  - Size of the blobs indicates the number of parameters



A. Canziani et al. „An Analysis of Deep Neural Network Models for Practical Applications". arXiv:1605.07678 2016

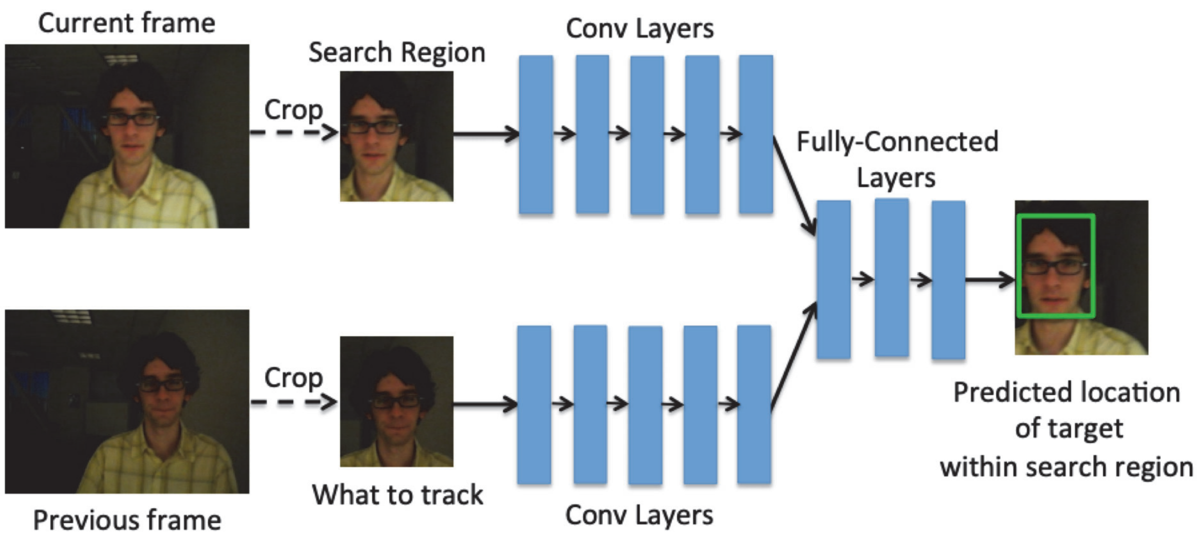
What can ML do for us?

- Regression problem: pose regression



What can ML do for us?

- Regression problem: bounding box regression



D. Held et al. „Learning to Track at 100 FPS with Deep Regression Networks“. ECCV 2016

What can ML do for us?

- Third type of problems

A  Classification: person, face, female

B  Classification: person, face, male

What can ML do for us?

- Third type of problems

A

Is it the same person?

B

What can ML do for us?

- Third type of problems: Similarity Learning

A 

B 

  - Comparison
  - Ranking

Similarity Learning: when and why?

- Application: unlocking your iPhone with your face

Training

Similarity Learning: when and why?

- Application: unlocking your iPhone with your face

A  YES

B  NO

Testing 

Can be solved as a classification problem

Similarity Learning: when and why?

- Application: face recognition system so students can enter the exam room without the need for ID check

Training

Person 1

Person 2

Person 3

Similarity Learning: when and why?

- Application: face recognition system so students can enter the exam room without the need for ID check

What is the problem with this approach?

Scalability – we need to retrain our model every time a new student is registered to the course

Similarity Learning: when and why?

- Application: face recognition system so students can enter the exam room without the need for ID check

Can we train one
model and use it every
year?

Similarity Learning: when and why?

- Learn a similarity function

A

B

Low similarity score

A

B

High similarity score

Similarity Learning: when and why?

- Learn a similarity function: testing

A

B

$$d(A, B) > \tau$$

Not the same person

Similarity Learning: when and why?

- Learn a similarity function

A

Same person $$d(A, B) < \tau$$

B

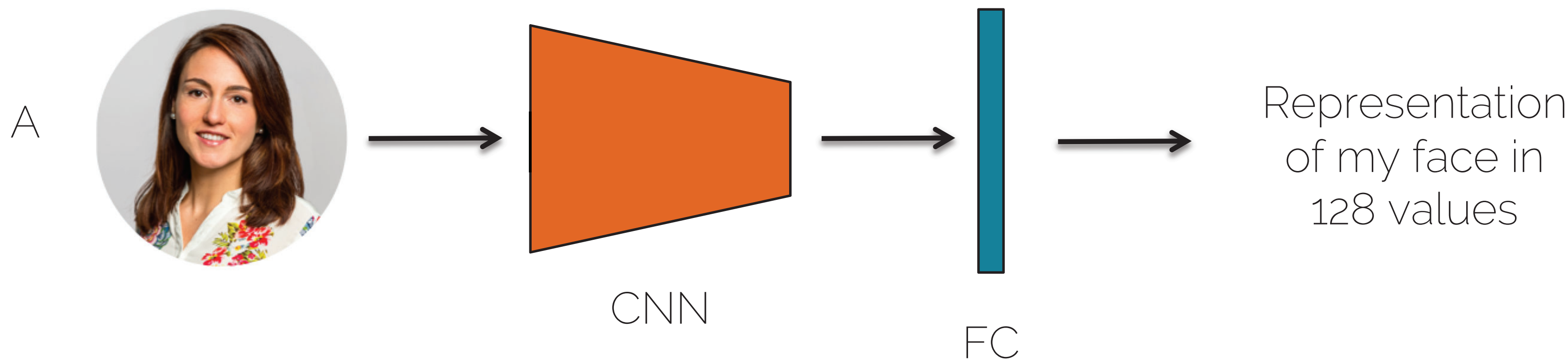
Similarity learning

- How do we train a network to learn similarity?

Siamese Neural Networks

Similarity learning

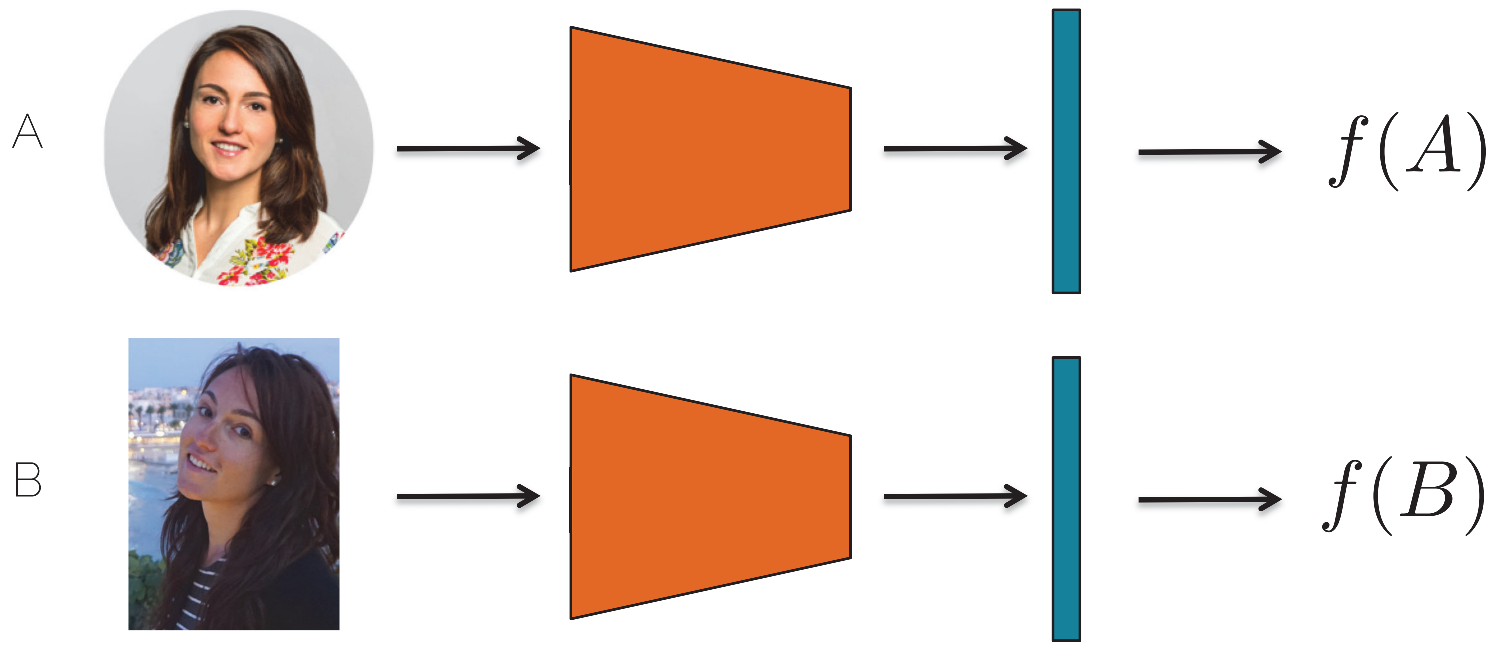
- How do we train a network to learn similarity?



Taigman et al. „DeepFace: closing the gap to human level performance". CVPR 2014

Similarity learning

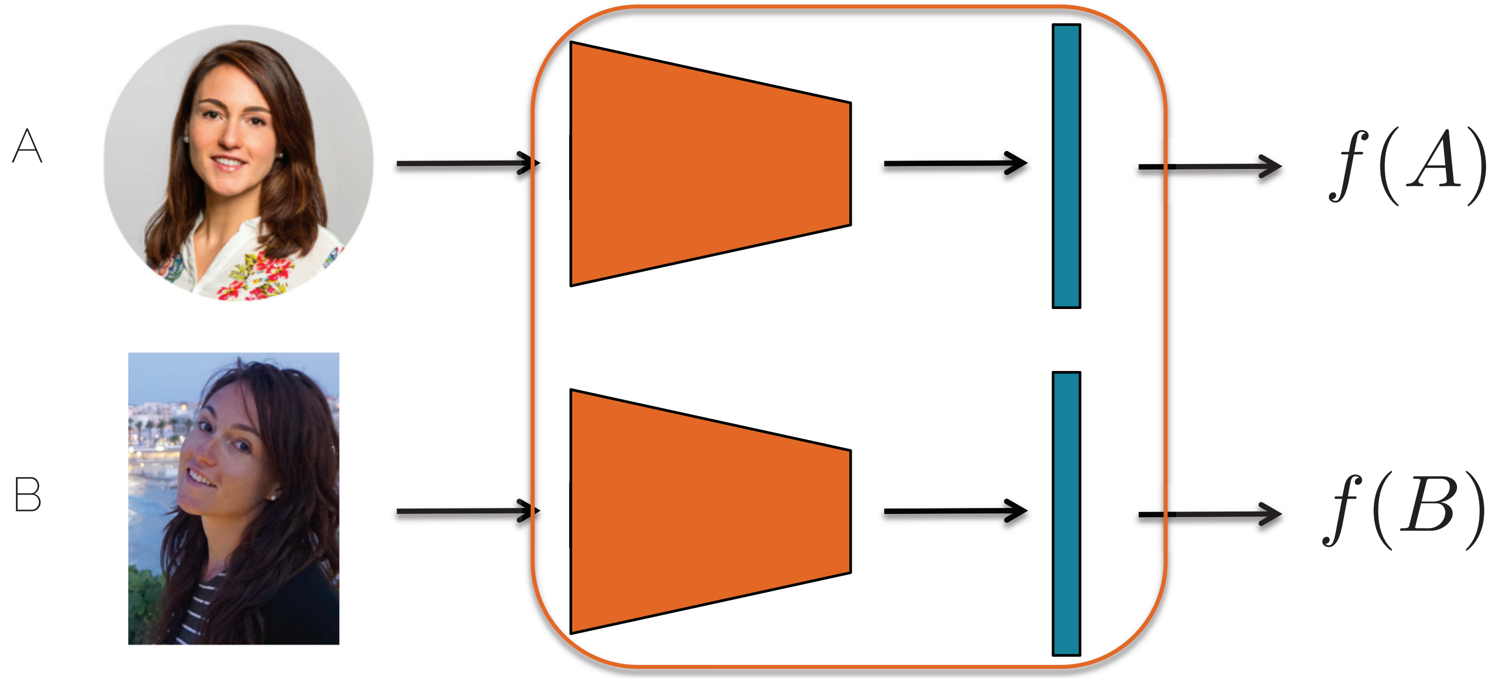
- How do we train a network to learn similarity?



Taigman et al. „DeepFace: closing the gap to human level performance“. CVPR 2014

Similarity learning

- Siamese network = shared weights



Taigman et al. „DeepFace: closing the gap to human level performance". CVPR 2014

Similarity learning

- Siamese network = shared weights
- We use the same network to obtain an encoding of the image $f(A)$
- To be done: compare the encodings

Taigman et al. „DeepFace: closing the gap to human level performance". CVPR 2014

Similarity learning

- Distance function $d(A, B) = ||f(A) - f(B)||^2$

- Training: learn the parameter such that
  - If A and B depict the same person, $d(A, B)$ is small
  - If A and B depict a different person, $d(A, B)$ is large

Taigman et al. „DeepFace: closing the gap to human level performance". CVPR 2014

Similarity learning

- Loss function for a positive pair:
  - If A and B depict the same person, $d(A, B)$ is small

$$\mathcal{L}(A, B) = ||f(A) - f(B)||^2$$

Similarity learning

- Loss function for a negative pair:
  - If A and B depict a different person, $d(A, B)$ is large
  - Better use a Hinge loss:

$$\mathcal{L}(A, B) = \max(0, m^2 - ||f(A) - f(B)||^2)$$

If two elements are already far away, do not spend energy in pulling them even further away

Similarity learning

- Contrastive loss:

$$\mathcal{L}(A, B) = y^*||f(A) - f(B)||^2 + (1 - y^*)max(0, m^2 - ||f(A) - f(B)||^2)$$

Positive pair,
reduce the distance
between the
elements

Negative pair,
brings the elements
further apart up to a
margin

Similarity learning

- Training the siamese networks
  - You can update the weights for each channel independently and then average them

- This loss function allows us to learn to bring positive pairs together and negative pairs apart

Triplet Loss

Triplet loss

- Triplet loss allows us to learn a ranking

Anchor (A)

Positive (P)

Negative (N)

We want: $$||f(A) - f(P)||^2 < ||f(A) - f(N)||^2$$

Schroff et al „FaceNet: a unified embedding for face recognition and clustering". CVPR 2015

Triplet loss

- Triplet loss allows us to learn a ranking

$$||f(A) - f(P)||^2 < ||f(A) - f(N)||^2$$

$$||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 < 0$$

$$||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 + m < 0$$

margin

Schroff et al „FaceNet: a unified embedding for face recognition and clustering". CVPR 2015

Triplet loss

- Triplet loss allows us to learn a ranking

$$||f(A) - f(P)||^2 < ||f(A) - f(N)||^2$$

$$||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 < 0$$

$$||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 + m < 0$$

$$\mathcal{L}(A, P, N) = max(0, ||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 + m)$$

Schroff et al „FaceNet: a unified embedding for face recognition and clustering". CVPR 2015

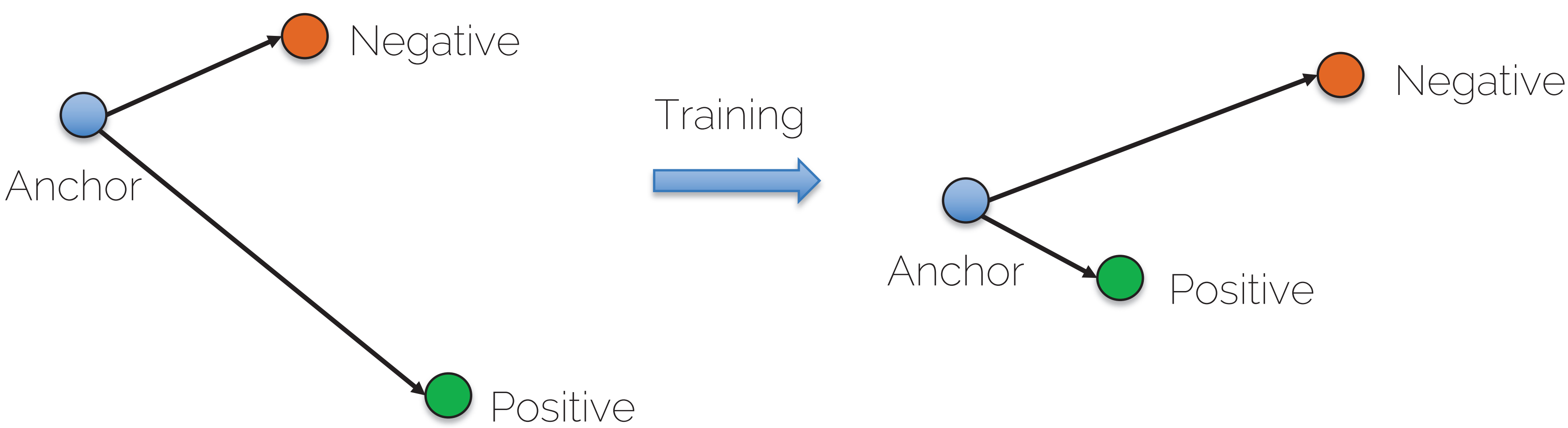
Triplet loss

- Hard negative mining: training with hard cases

$$\mathcal{L}(A,P,N) = max(0, ||f(A) - f(P)||^2 - ||f(A) - f(N)||^2 + m)$$

- Train for a few epochs
- Choose the hard cases where $d(A,P) \approx d(A,N)$
- Train with those to refine the distance learned

Triplet loss



Triplet loss: test time

- Just do nearest neighbor search!

Triplet Loss Challenges

- Random sampling does not work - the number of possible triplets is $O(n^3)$ so the network would need to be trained for a very long time.

- Even with hard negative mining, there is the risk of being stuck in local minima.